Exercise Sheet 17

All modules are finite-dimensional over $\mathbb{C}$. Exercises 17.1–17.4 are calculations, two of which expose common errors. Exercise 17.5 reconstructs the limit in the proof of the dimension formula. Exercises 17.6–17.7 apply the methods to $\mathfrak{sl}_3$.

For type C_2 use the following data. Let $\mathfrak{g} = \mathfrak{sp}_4(\mathbb{C})$, with Cartan subalgebra the diagonal matrices $\operatorname{diag}(a_1, a_2, -a_1, -a_2)$, and let ε_i read a_i. Then

$$R_+ = \{\varepsilon_1 - \varepsilon_2,\ \varepsilon_1 + \varepsilon_2,\ 2\varepsilon_1,\ 2\varepsilon_2\}, \qquad \alpha_1 = \varepsilon_1 - \varepsilon_2, \quad \alpha_2 = 2\varepsilon_2.$$

The Weyl group acts on $\mu = \mu_1\varepsilon_1 + \mu_2\varepsilon_2$ by all permutations and sign changes of (μ_1, μ_2). A swap or a single sign change is a reflection. For the form, use the dot product in ε-coordinates. It is W-invariant because signed permutations preserve the dot product. The coroot pairings do not depend on a rescaling of this form.

Exercise 17.1 (The C_2 dimension formula).

(i) Compute $\langle \mu, \alpha^\vee \rangle$ for each $\alpha \in R_+$. Find the Cartan matrix in the convention $a_{ij} = \alpha_j(h_i)$, the fundamental weights, and ρ.

(ii) For integers $a, b \geq 0$, prove that

$$\dim L(a\omega_1 + b\omega_2) = \frac{(a+1)(b+1)(a+2b+3)(a+b+2)}{6}.$$

(iii) Tabulate the dimensions for $(a, b) = (1, 0), (0, 1), (2, 0), (1, 1), (0, 2)$. Check two values against modules you know.

Exercise 17.2 (Roots are not coroots). A student applies the simply-laced shortcut $\langle \lambda + \rho, \alpha^\vee \rangle = \sum_i c_i(m_i + 1)$ for $\alpha = \sum_i c_i\alpha_i$ in type C_2, with $\lambda = a\omega_1 + b\omega_2$ and integers $a, b \geq 0$.

(i) Write $\varepsilon_1 + \varepsilon_2$ and $2\varepsilon_1$ in terms of simple roots. Find the student's formula $F(a, b)$.

(ii) Compare $F(1, 0)$ and $F(2, 0)$ with the correct values. Express F in terms of the correct formula $D(a, b)$ of Exercise 17.1.

(iii) Find the coroot coefficients of $(\varepsilon_1 + \varepsilon_2)^\vee$ and $(2\varepsilon_1)^\vee$ by comparing pairings.

(iv) The adjoint module of $\mathfrak{sp}_4$ has how many distinct weights? Why is that not its dimension?

Exercise 17.3 (Kostant's formula in C_2).

(i) Let $u, v \in \mathbb{Z}$. Show that $u\varepsilon_1 + v\varepsilon_2$ lies in Q_+ exactly when $u \geq 0$, $u + v \geq 0$ and $u + v$ is even.

(ii) Compute $p(2\alpha_1 + \alpha_2)$ and $p(2\alpha_1)$.

(iii) Use Kostant's formula to compute $m_{2\omega_1}(0)$. Check the result against Exercise 17.1.

Exercise 17.4 (Forgetting the ρ-shift). Let $\mathfrak{g} = \mathfrak{sl}_2$, $R_+ = \{\alpha\}$, $\lambda = m\omega$ with $m \in \mathbb{Z}_{\geq 0}$, and $\mu = \lambda - k\alpha$ with $k \in \mathbb{Z}$.

(i) Evaluate Kostant's formula and recover the weights of V_m.

(ii) A student uses $\sum_w \det(w)\, p(w\lambda - \mu)$ instead. Show that, for $m \geq 1$, the result contradicts W-invariance. What goes wrong for $m = 0$?

Exercise 17.5 (Why the direction must be regular).

(i) For $\nu \in P$, prove that $\Phi_\nu(A_\rho) \neq 0$ exactly when ν is regular.

(ii) For $\mathfrak{sl}_3$, show $\Phi_{\omega_1}(A_\rho) = 0$. Show also that $A_{\omega_1} = 0$, and deduce that Φ_{ω_1} applied to the character formula gives $0 = 0$ for every $\lambda \in P_+$.

(iii) For $\mathfrak{sl}_2$, let $\lambda = m\omega$ with $m \in \mathbb{Z}_{\geq 0}$ and $(\alpha, \alpha) = 2$. Compute $\Phi_\rho(A_\rho)$ and $\Phi_\rho(A_{\lambda+\rho})$ to order t, and recover $\dim V_m$.

Exercise 17.6 (Character calculations in $\mathfrak{sl}_3$). Let $\mathfrak{g} = \mathfrak{sl}_3$ and $V = \mathbb{C}^3$.

(i) List the dominant weights of $V \otimes V$ with multiplicities.

(ii) Use Kostant's formula to find $m_{2\omega_1}(\omega_2)$. Decompose $V \otimes V$.

(iii) Identify $\Lambda^2 V$ and $\mathrm{Sym}^2 V$ as irreducible modules.

(iv) For every integer $n \geq 0$, compute $m_{n\rho}(0)$ for $\mathfrak{sl}_3$. Check $n = 2$.

Exercise 17.7 (Small $\mathfrak{sl}_3$ dimensions).

(i) Find all nonnegative integer pairs (p, q) with $\dim L(p\omega_1 + q\omega_2) \leq 10$.

(ii) Find all irreducible $\mathfrak{sl}_3$-modules of dimension 15. Which pairs are dual to each other?